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Paleomagnetic results in support of a model for the origin of the Asturian arc

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ABSTRACT

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Paleomagnetic directions have been determined at 17 Cambrian and 28 Carboniferous limestone sites distributed along the Asturian arc of the Hercynian fold belt in northwestern Spain. The stable vectors are carried by hematite in all cases, and also by magnetite in some grey Carboniferous limestones. A secondary component of magnetization is usually removed well below 400°C, and the higher-temperature component is taken to be the characteristic direction. The structure at each site was carefully evaluated in order to make optimum local tectonic corrections. In addition the data must be corrected for rotations about a vertical axis to allow for a phase of radial folding superimposed on earlier structures. The corrected paleomagnetic declinations are found to vary systematically along the arc of the fold belt. Consequently, paleomagnetic data from the Asturian arc should not be included in compilations used for the construction of a Paleozoic apparent polar wander path.

The paleomagnetic data allow us to distinguish between different tectonic models for the evolution of the Asturian arc. A two-stage model for the development of the present curvature is favored. Part of the curvature appears to be primary, preceding the Hercynian deformation. The first stage of the model involved rotations of thrust sheets during their emplacement, producing a more tightly curved arc than the original form. In a second stage, the development of radial folds further tightened the curvature of the arc. Both stages result in clockwise rotations in the north and anticlockwise rotations in the south.

Introduction

Arcs are first-order structures in convergent plate boundaries. For example, island arcs are found as primary structures in present-day subduction zones, and arcuate mountain belts are produced in continent–continent collisions. Modern as well as older fold belts also show curved features which have been interpreted variously as primary, secondary or composite features. A secondary origin of arcs by bending of a straight orogenic belt was postulated by Carey (1955), who named these features oroclines. However, the total or partial secondary origin can also be explained by other mechanisms (Marshak, 1988), such as the indentation of rigid promontories along irregular colliding continental margins which influences the deformational history, structural trends and strain distribution of the resulting fold belt.

In the Cantabrian mountains of northwestern Spain, the Asturian arc forms the core of a broader structure—the Ibero-Armorican arc—which connects the northwest-trending Iberian structures with the east-trending Armorican structures (Fig. 1). The Asturian arc was first referred to by Suess (1885–1909), although it was already shown in older maps (Schulz, 1858). More than a century later de Sitter (1965) attempted the first structural interpretation in which he

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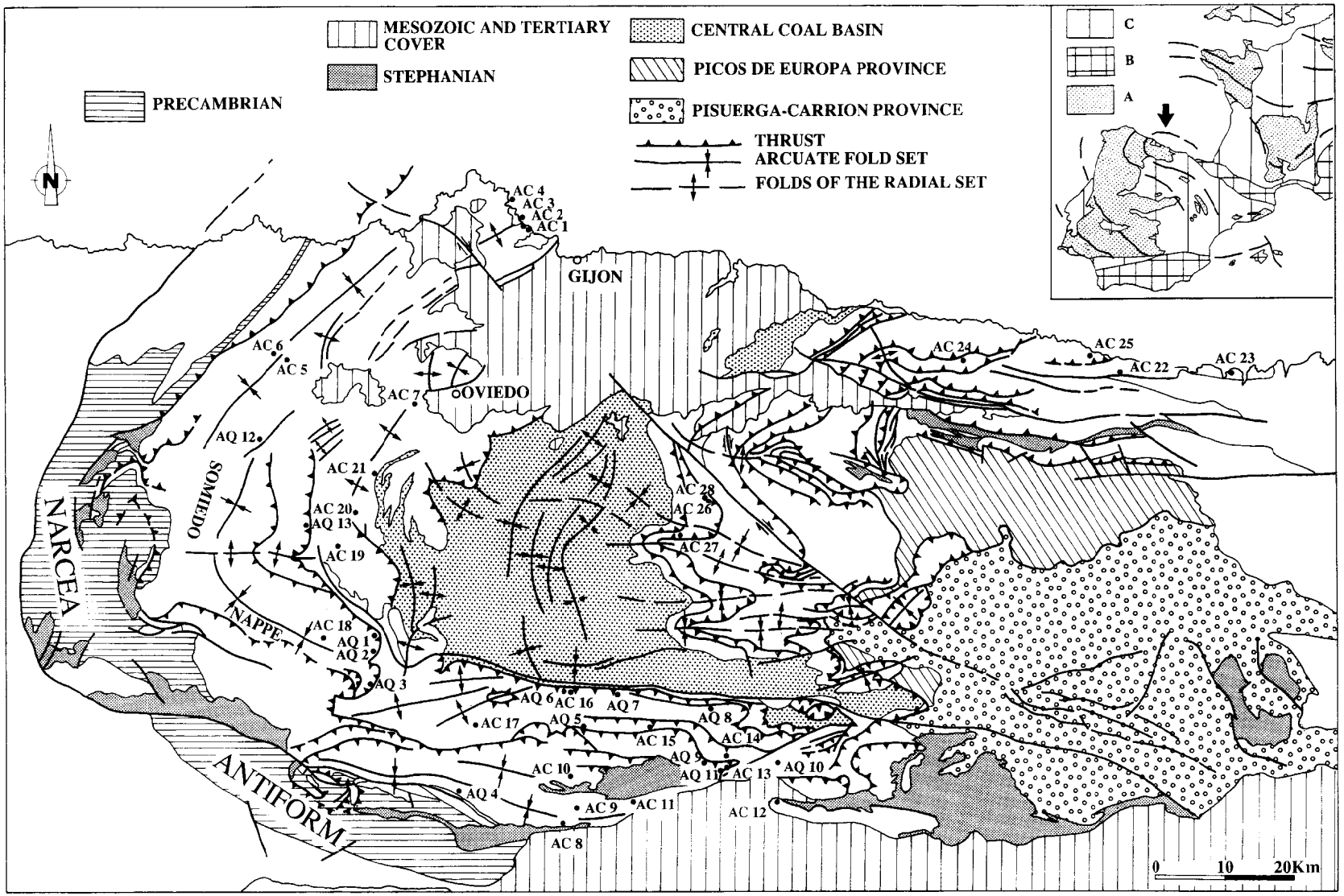


Fig. 1. Schematic tectonic map of the Asturian arc with paleomagnetic site localities. Cambrian sites are labeled *AQ* and Carboniferous sites *AC*. Inset: Ibero-Armorican arc; arrow indicates study area. *A* = Hercynian Massif; *B* = Alpine fold belt; *C* = Mesozoic and Tertiary platform cover.

explained the arc as resulting from two deformation phases, which gave the east–west and north–south segments, respectively. However, the continuity of many structures along the arc precludes de Sitter's interpretation.

In a study of strain variations in arcuate fold belts, Ries and Shackleton (1976) investigated the origin of curvature of the Ibero-Armorican arc in the Iberian Peninsula. The authors analyzed the different theoretical strain patterns which would result from different arc evolutions. The finite extension in the outer part of this arc increases outwardly and the stretching lineation plunges at a low angle parallel to the main fold axes. In the inner part of the arc the finite stretching lineation changes to a down-dip direction and fans around the arc, perpendicular to the main fold axes. From the observed strain pattern, Ries and Shackleton (1976) concluded that the Ibero-Armorican arc is similar to a buckle fold with tangential longitudinal strain, and hence is secondary in origin.

Julivert (1971a) and Julivert and Marcos (1973) showed that a second fold system overprints the main folds and thrusts which define the Asturian arc. These folds are radial, extending outwards from the center of the Asturian arc normal to the main fold axes. They concluded that the arc was primary in origin but was tightened during the latest stage of its development by the radial folds. Matte and Ribeiro (1975), Matte (1986) and Julivert and Martinez (1987) later explained the tightening of the arc by the indentation produced by relative motion of its core parallel to its flanks.

Julivert and Arboleya (1984, 1986) analyzed kinematic indicators in order to understand thrust emplacement of individual nappe sheets and their bearing on the evolution of the Asturian arc. These investigations further supported a composite origin of the arc. A primary arc was being tightened as deformation progressed, first by rotations of thrust sheets during their emplacement, and later by the generation of the radial set of folds. Pérez-Estaún et al. (1988) postulated a similar model for the origin of the arc, but their model differs from that of Julivert and Arboleya (1984, 1986) in the location of the rotation centers.

A model proposed by Ries and Shackleton (1976) implies that anticlockwise rotation should have occurred over the entire Variscan arc. The models of Matte and Ribeiro (1975), Matte (1986) and Julivert and Arboleya (1984, 1986) imply that the northern side of the Variscan arc would have undergone clockwise rotation and the southern side of the arc would have undergone anticlockwise rotation. The model presented by Pérez-Estaún et al. (1988) predicts clockwise rotation of the outer thrust sheets around the arc.

In order to clarify which model represents the development of the Asturian arc, more information is necessary about: (1) the primary curvature of the arc; (2) the kinematics of the thrust sheets; and (3) how the Asturian arc fits into the Hercynian fold belt with respect to lithospheric plates and their motions. Paleomagnetic directions provide an additional parameter that may help to distinguish which model best describes the formation of the Ibero-Armorican arc, and several paleomagnetic studies have been conducted in the Asturian arc with this goal.

Ries et al. (1980) conducted the first paleomagnetic study in the Asturian arc. The results of this investigation were discussed and re-evaluated by Lowrie and Hirt (1986). The majority of the sites were not demagnetized, and those which were treated with thermal demagnetization often had a large within-site scatter ($\alpha_{95} > 15°$). Ten sites were found suitable to test the origin of curvature of the arc. Although Ries et al. (1980) provided little information about the quality of the demagnetization or about the structural corrections which were applied to the sites, the site mean directions suggested that the curvature of the arc was secondary, due to the bending of an originally straight fold belt.

The Asturian arc has been investigated in other paleomagnetic studies. In the Villasecino area (located between sites AQ3 and AC17, Fig. 1), where the trend of the fold axes changes from east–west to north–south, Bonhommet et al. (1981) applied detailed structural corrections to account for the two phases of folding. The characteristic remanent magnetization (ChRM) directions are pre-tectonic in origin, but the paleopole defined for the region is rotated 50–65° anti-

clockwise about a vertical axis from the expected Carboniferous pole for Europe after accounting for the opening of the Bay of Biscay. The authors concluded that the curvature of the arc was secondary in origin, but they could not discriminate between models of arc formation.

Ordovician and Carboniferous lithologies were sampled in the Cabo de Peñas (Perroud, 1983). The Ordovician directions were in good agreement with other Lower Paleozoic data from the Armorican plate. The Carboniferous directions were interpreted to be either pre- or synfolding in age. Perroud (1983) suggested that an additional "strike correction", a rotation about a vertical axis, was necessary to account for the later folding phase.

Perroud (1985) sampled the Cambrian Láncara Formation, the Devonian San Pedro Formation and the Carboniferous Alba Formation. The ChRM directions were well grouped within the majority of the sites, but the between-site scatter was high, both in declination and inclination. Perroud (1985) concluded that some of the paleomagnetic directions had a synfolding origin, but that directions from the San Pedro and Alba formations were pre-tectonic, probably of diagenetic origin. The declinations grossly followed the shape of the arc, but they could not be correlated precisely with structural directions.

In this new study we combine thorough analysis of the magnetization vector with detailed tectonic corrections at both the site and regional level.

Geological setting

The Hercynian fold belt in northwestern Spain exhibits a zonation from high-grade metamorphic areas in the west to a non-metamorphic zone in the east (Julivert and Martinez, 1987). The Asturian arc is a thin-skinned fold–thrust belt, which is concave towards the foreland and whose structures face inwards. The structural trend varies by about 180° around the arc. The western boundary of the Asturian arc is the Narcea antiform (Fig. 1).

The geology of the area is fairly well known, and maps on the scale 1:50,000 are available for the whole area. The most significant contributions to mapping the area have been achieved by the Oviedo and Leiden schools. Their results have been published in journals of these institutions and as sheets of the Geological Map of Spain. Directly relevant to our project are studies in the southern part of the Cantabrian arc by Rupke (1967), Evers (1967), Sjerp (1967) and Van den Bosch (1969), compiled by de Sitter and Boschma (1966) and Savage and Boschma (1980). Studies by Julivert et al. (1968, 1973, 1977), Martinez-Alvarez et al. (1975), Pello (1976) and Crespo (1982) are applicable to the central and northern parts of the arc.

The structural pattern of the zone results from the interference between (1) thrust sheets, (2) arcuate folds, describing the arc, and (3) radial folds which cut the first fold set (Julivert, 1971b; Julivert and Marcos 1973). The thrust sheets and the associated arcuate fold set developed first, with detachment occurring near or at the base of the Paleozoic sequence in the Early to Middle Cambrian Láncara Formation. Alonso (1987) and Pérez-Estaún et al. (1988) suggest that the folds are partially a consequence of duplexes and antiformal stacks. However, most folds of the arcuate set are believed to have formed where the thrust sheets die out, since this would accommodate their displacement laterally into a fold structure (Julivert and Arboleya, 1984, 1986).

As deformation progressed, the growth of the thrust sheets and the irregularities produced by hanging-wall ramp anticlines, duplexes and antiformal stacks, made motion difficult. The existing folds were tightened and the thrust sheets began to fold. The newly generated folds were essentially nucleated in pre-existing structures and were, therefore, longitudinal with respect to the older folds and thrusts. The radial folds superposed on the above structures are best defined in the core of the Asturian arc and vanish towards its outer peripheries.

Lithology

The Paleozoic sequence in the Cantabrian zone can be divided on the basis of lithology into two parts.

TABLE 1

Locations of paleomagnetic sites

Site [a]	Locality	Longitude (W)	Latitude (N)	Bedding (Azimuth/Dip)	Remarks [b]
AQ 1	Puerto Ventana	6°0′ 47″	43°3′ 37″	222/62	FA 239/62
AQ 2	Torrestío	6°1′ 3″	43°1′ 55″	290/72	FA 239/62
AQ 3	La Majúa	6°1′ 40″	42°59′20″	303/50	FA 289/53
AQ 4	Barrios de Luna	5°51″ 29″	42°50′37″	40/72	FA 114/34
AQ 5	Villamanin	5°39′ 8	42°56′6″	21/66	Overturned
AQ 6	Camplongo	5°41′22″	42°59′12″	173/80	
AQ 7	Pontedo	5°33′21′	42°58′35″	26/65	Overturned
AQ 8	Tolibia de Abajo	5°24′15″	42°58′00″	359/80	Overturned
AQ 9	Valdorria	5°24′53″	42°53′34″	323/78	Overturned
AQ 10	Cerecedo	5°18′30′	42°53′21″	337/50	Overturned
AQ 11	Montuerto	5°23′38′	42°53′10″	322/41	
AQ 12	Corias	6°12′25′	43°18′45″	320/81	
AQ 13	Villamayor	6°18′12′	43°11′57″	265/54	
AC 1	Carranques	5°44′37″	43°34′43″	334/57	
AC 2	Carranques	5°44′37″	43°34′43″	328/60	
AC 3	Carranques	5°44′44″	43°35′ 5″	209/46	
AC 4	Carranques	5°46′11″	43°36′ 3′	126/80	Overturned
AC 5	Figares	6°10′44″	43°23′41″	125/45 [c]	FA 213/1
AC 6	Villacarisme	5°12′10″	43°24′57″	303/68 [d]	FA 213/1
AC 7	S. Pedro de Nora	5°57′44″	43°22′15″	314/44	
AC 8	Olleros de Alba	5°42′ 7″	42°48′14″	7/88	
AC 9	Buen Suceso	5°39′12″	42°49′55″	358/68	Overturned FA 297/30
AC 10	Pola de Gordón	5°40′ 7″	42°51′54″	1/88	Overturned
AC 11	Matallana	5°31′17″	42°50′48″	354/77	Overturned
AC 12	S. Adrián	5°18′21″	42°50′31″	221/51	
AC 13	Montuerto	5°23′15″	42°53′ 2″	313/83	
AC 14	Caldas de Nocedo	5°23′47″	42°54′ 9″	358/83	Overturned
AC 15	Getino	5°32′46″	42°56′20″	357/77	Overturned
AC 16	Millaró	5°38′58″	42°58′35″	17/87	Overturned
AC 17	Caldas de Luna	5°50′37″	42°56′20″	171/76	
AC 18	Iron Mine	6°6′ 22″	43°3′20″	180/31	
AC 19	Teverga	6°4′ 47″	43°11′ 6″	43/78	Overturned
AC 20	Caranga	6°2′ 3″	43°13′57″	57/73	
AC 21	Santo Adriano	5°59′15″	43°17′58″	130/64	
AC 22	Rio Purón	4°41′42″	43°23′46″	187/80	
AC 23	Tinamayor	4°30′39″	43°23′34″	0/60	
AC 24	El Torno	4°58′42″	43°24′26″	96/30	
AC 25	Llanes	4°45′35″	43°24′24″	178/68	
AC 26	Rioseco	5°26′13″	43°13′40″	[e]	
AC 27	Rioseco	5°28′16″	43°11′17″	[e]	
AC 28	Rioseco	5°26′13″	43°13′40″	[e]	

[a] AQ designates sites of Cambrian age and AC designates sites of Carboniferous age.

[b] FA indicates azimuth and plunge of fold axis.

[c] Local bedding and direction were first corrected for average bedding at site, 125/61.

[d] Local bedding and direction were first corrected for average bedding at site, 303/70.

[e] See Julivert and Arboleya (1986) for structural correction applied.

Pre-Carboniferous rocks form a wedge-shaped, pretectonic, shallow-water platform sequence which thickens towards the convex part of the arc. From Cambrian to Silurian the sequence is essentially terrigenous. The only carbonate unit is the Cambrian Láncara Formation which consists of an Early Cambrian dolostone and fenestral limestone member, and a Middle Cambrian nodular limestone member (Zamarreño, 1972, 1975). The Devonian consists of an alternating sequence of terrigenous and carbonate formations.

During the Carboniferous, between Late Visean and Middle Namurian, a major change in sedimentation occurred and a thick terrigenous sequence was deposited. This sequence marks the beginning of the Hercynian orogeny and a change from syntectonic to post-tectonic conditions (Julivert, 1978; Marcos and Pulgar, 1982).

Between the two sequences, a thin (20–50 cm) condensed succession is found with an age ranging from Tournaisian to the end of the Visean. This sequence consists of several very thin formations which include the Alba Formation. This formation is predominantly a red, nodular limestone that outcrops throughout the Asturian arc with a constant thickness and lithology. In several areas the Alba Formation is succeeded by a turbiditic sequence; however, in general it is overlain by the Namurian-age Barcaliente Formation, a black/grey limestone.

Strain is not significant in the Asturian arc and cleavage is only locally developed. Stylolites are present in the Cambrian and Carboniferous nodular limestones. These are generally diagenetic features and are parallel to the bedding. When cleavage is locally developed, it is generally of pressure-solution origin.

Sample collection and treatment

Three lithologies were sampled in this new study; the upper member of the Cambrian Láncara Formation and the Carboniferous Alba and Barcaliente Formations. The Láncara Formation is largely exposed throughout the arc. However, the facies of the upper member are not uniform and two provinces can be distinguished (Zamarreño, 1972, 1975). East of the Central Coal basin, the upper member is made up of nodular, glauconitic limestones which are very rich in echinoderm debris. To the south and west of the basin, the upper member consists of a red, fine-grained nodular limestone, devoid of echinoderm debris. This latter facies was sampled. The sites are concentrated in the south and west because the Láncara Formation does not crop out where the tectonic style changes from thrusting to folding. Approximately 300 specimens were collected with a portable drill from 17 sites, 13 of which gave usable paleomagnetic results (Fig. 1, Table 1). Data from 4 sites in the Elsa nappe were not used, because it was not possible to correct them adequately for the complex local structures.

The Alba and Barcaliente Formations have uniform facies throughout the entire Cantabrian zone. Sampling sites were distributed along the whole arc in these two formations. Sampling in the Barcaliente Formation was generally limited to the lowermost part, i.e. within a few metres of its contact with the Alba formation. Over 600 specimens were drilled from 28 sites around the arc in these formations (Fig. 1, Table 1).

All remanent magnetizations were measured with a three-axis ScT cryogenic magnetometer with a sensitivity equivalent to 8×10^{-6} A/m for a standard cylindrical specimen 2.54 cm in diameter and 2.25 cm high. Specimens were thermally demagnetized with a Schonstedt thermal demagnetizer with a rest field of approximately 10 nT. The bulk susceptibility of thermally treated specimens was monitored after each heating step with a KLY-1 bulk susceptibility bridge.

Rock magnetism

The acquisition of isothermal remanent magnetization (IRM) was observed in fields up to 5.0 T using an Oxford cryogenic magnet in several specimens of each lithology. The presence of ferrimagnetic minerals with different coercivities may be analyzed by deliberately magnetizing the different coercivity fractions in orthogonal directions, and observing their behavior during subsequent thermal demagnetization (Lowrie, 1990). In

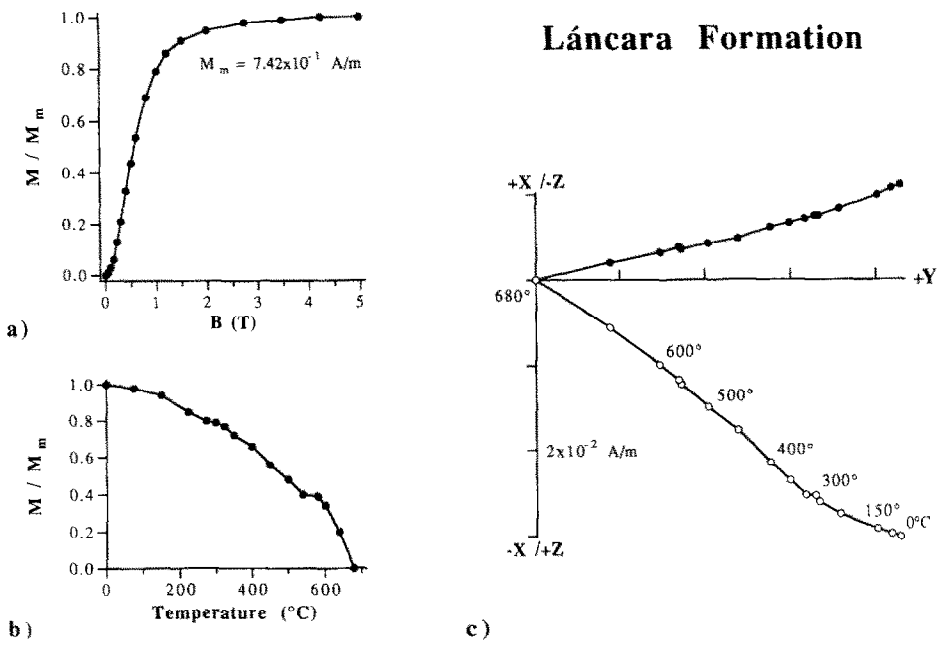


Fig. 2. (a) Acquisition of IRM of a representative sample from the Láncara Formation. (b) Normalized intensity plot showing the intensity variation. (c) Orthogonal vector diagram showing the directional behavior during thermal demagnetization. For this and subsequent figures, closed symbols on the orthogonal vector diagram represent vector end points projected on the horizontal plane; open symbols are for the vertical plane.

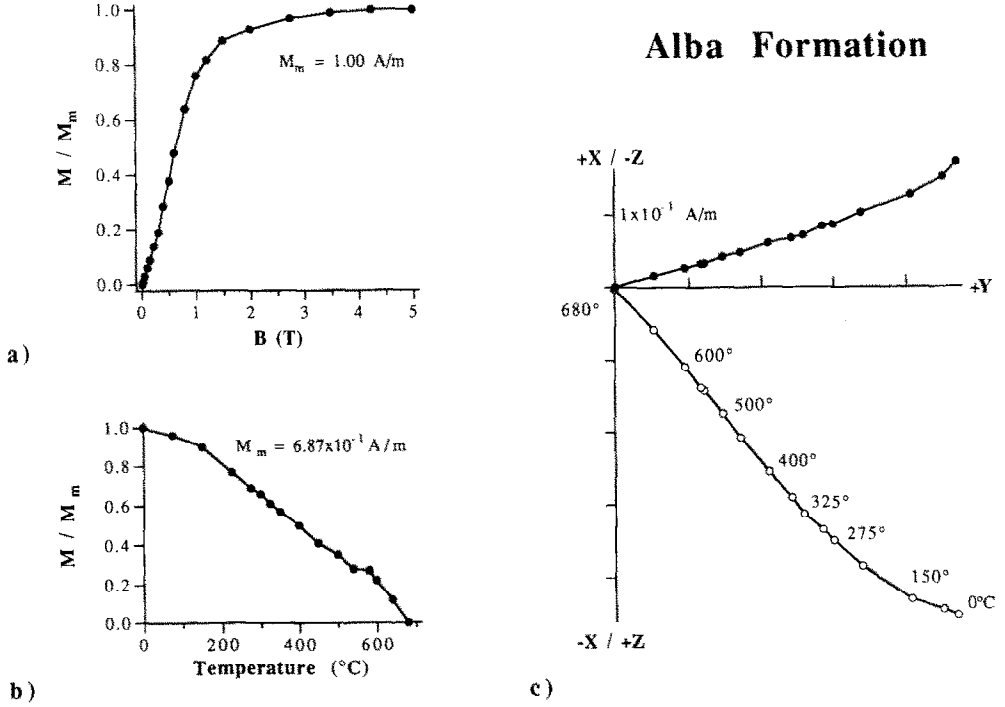


Fig. 3. (a) Acquisition of IRM of a representative sample from the Alba Formation. (b) Normalized intensity plot showing the intensity variation. (c) Orthogonal vector diagram showing the directional behavior during thermal demagnetization.

the present case, the sample was first magnetized along the z-axis (cylinder axis) in a field of 5.0 T. Next, an 0.5 T field was applied along the y-axis of the sample, and finally an 0.1 T field was applied along the x-axis. The samples were subsequently demagnetized thermally in order to identify the ferrimagnetic minerals in the rock.

IRM acquisition curves for the Láncara limestone are slightly concave upward in low fields, indicating a lack of minerals with extremely low coercivity. In higher fields the curve is dominated by a high coercivity component and saturation IRM is not reached by 5.0 T (Fig. 2a). Thermal demagnetization shows a distributed unblocking of the IRM (Fig. 2b), revealed in more detail when analyzed with vector diagrams (Zijderveld, 1967). Apart from minor differences, each of the three coercivity fractions demagnetizes smoothly up to 660°C (Fig. 2b, c). These coercivity and thermal unblocking characteristics suggest that hematite is the main magnetic mineral in the Láncara limestone. The distributed unblocking temperature spectrum, and uniform demagnetization of the three coercivity fractions suggests that the hematite has a wide range of grain sizes.

The Carboniferous sites include both a red "griotte" (nodular) limestone and a grey limestone. The features of the IRM acquisition and subsequent thermal demagnetization of the red griotte resemble closely those of the Láncara limestone (Fig. 3a–c). The red griotte specimens are governed by a high-coercivity, high unblocking temperature mineral, which we interpret to be hematite.

The acquisition of IRM and thermal demagnetization of the composite IRM in the grey limestones of the Barcaliente formation display different behavior than the red limestones. Both low and high coercivity components are indicated by the IRM curves (Fig. 4a). The thermal demagnetization indicates that more than one ferromagnetic mineral is present in the rocks (Fig. 4b, c). A high coercivity component (> 0.5 T) is removed between room temperature and 150°C. Goethite is most likely responsible for this high-coercivity, low-temperature component of magnetization. Between 150°C and 450°C magnetization changes on the low, intermediate and high coercivity axes are observed, which are dominated by the component softer than 0.1 T. Magnetite is

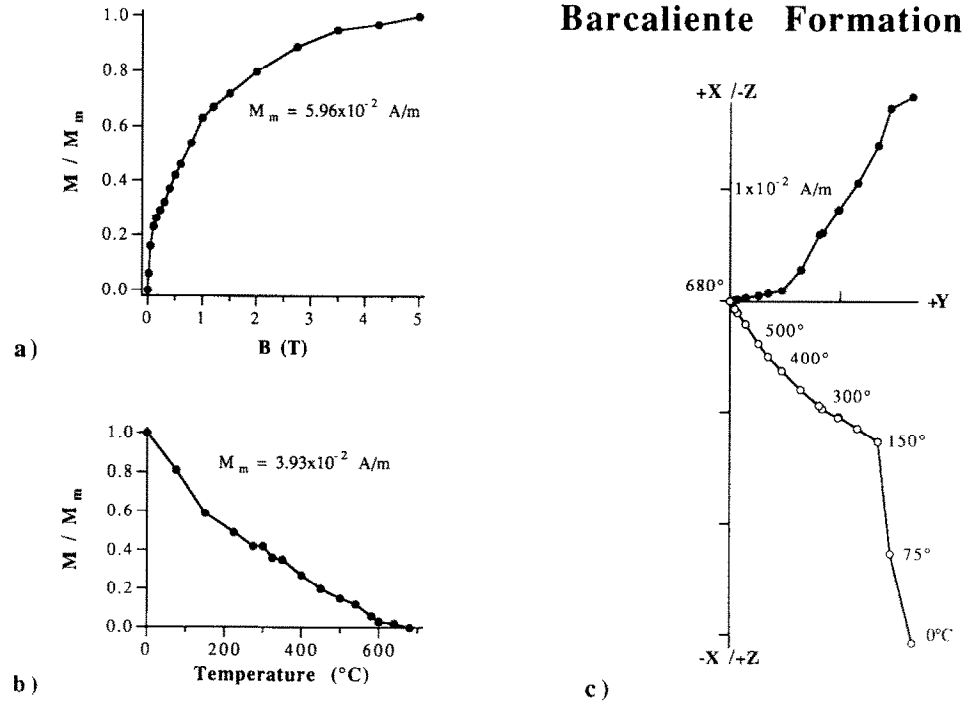


Fig. 4. (a) Acquisition of IRM of a representative sample from the Barcaliente Formation. (b) Normalized intensity plot showing the intensity variation. (c) Orthogonal vector diagram showing the directional behavior during thermal demagnetization.

most likely responsible for this low-coercivity component of magnetization. A further high-coercivity component, thermally removed between 450°C and 660°C is attributed to hematite, which also accounts for the harder fractions removed above 150°C.

Paleomagnetic results

The components of natural remanent magnetization were determined from orthogonal vector diagrams (Zijderveld, 1967) of the thermal demagnetization behavior. Vectors were defined by least squares fitting of linear segments in the vertical and horizontal planes of projection.

Cambrian sites

The NRM intensities of the Láncara formation were relatively strong ($1-5 \times 10^{-3}$ A/m). Specimens were thermally demagnetized and generally revealed two components of magnetization after removing a spurious direction below 100–150°C. A further component, which was usually defined between 100 and 360°C, presumed to be of secondary origin, had a similar direction to the high-temperature component defined between 400 and 660°C (Fig. 5a), which we regard as the characteristic remanent magnetization (ChRM). The secondary magnetization was less well-grouped than the ChRM directions, which were very well grouped within each site and always gave α_{95} less than 8° (Table 2).

Carboniferous sites

The NRM intensities in the Carboniferous sites were dependent on lithology. The griotte limestones had an initial intensity in the range $1-5 \times 10^{-3}$ A/m. As in the Láncara formation, two

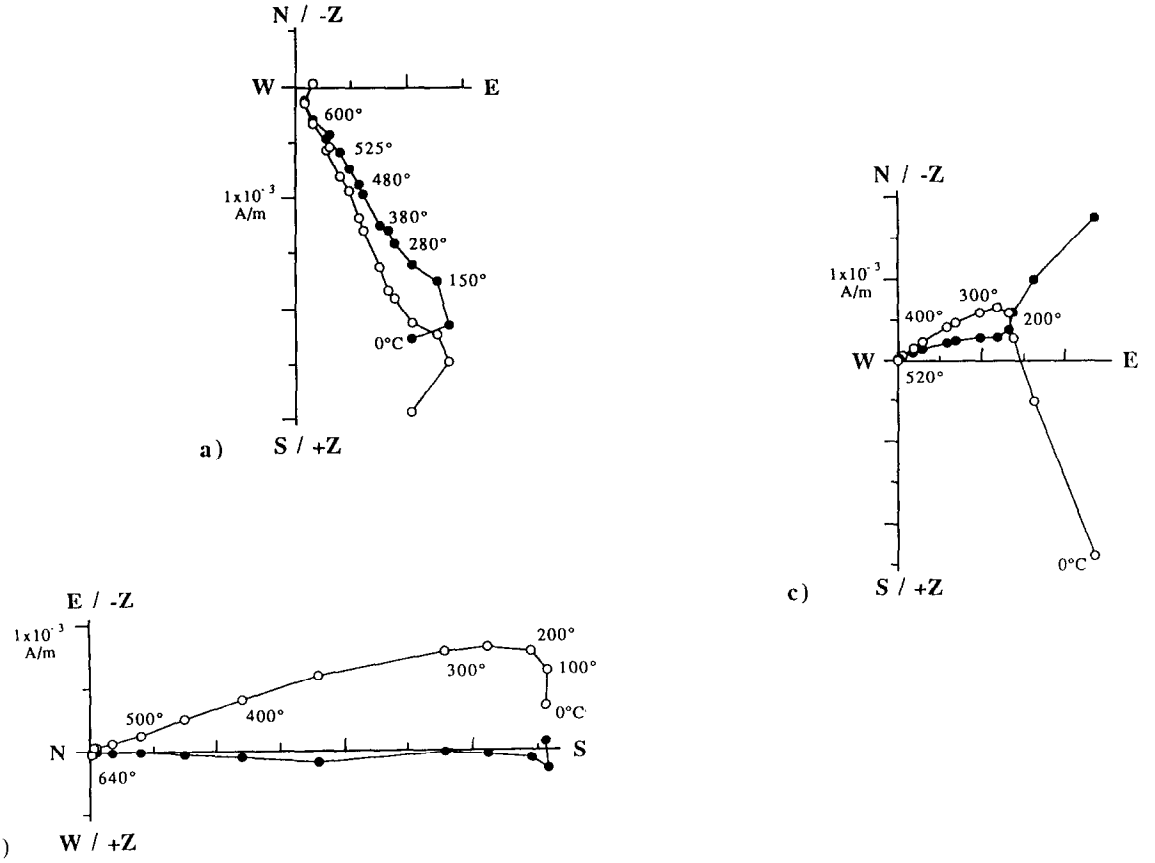


Fig. 5. Vector diagrams of thermally demagnetized specimens from (a) Láncara Formation, (b) Alba Formation and (c) Barcaliente Formation.

TABLE 2

Site mean directions from the Cambrian Lancara Formation

Site	N [a]	In situ		After local tectonic correction		After removal of radial folds		
		D	I	D	I	D	I	α_{95}
Group I								
AQ12	9	154.0	−38.0	154.5	41.0	144.5	41.0	4.2
Group II								
AQ13	8	122.0	4.0	141.0	43.0	141.0	43.0	3.8
Group III								
AQ1	8	100.0	−3.5	122.0	26.0	130.0	26.0	3.7
AQ2	8	147.0	−29.0	117.0	34.0	125.0	34.0	3.5
AQ3	8	132.5	−15.5	122.0	39.0	130.0	39.0	6.6
AQ4	8	149.0	46.0	109.0	29.0	126.0	29.0	4.0
AQ5	7	84.0	8.5	130.0	20.0	147.0	20.0	7.8
AQ6	8	98.5	10.0	93.5	−8.5 [b]			6.6
AQ7	9	93.5	−2.5	136.5	26.0	153.5	26.0	4.0
AQ8	8	70.0	22.0	115.0	13.5	134.0	13.5	7.3
AQ9	9	132.5	−38.0	334.0	−39.0	351.0	−38.0	4.7
AQ10	8	80.5	−53.5	16.0	24.0 [b]			2.7
AQ11	9	143.0	−24.5	143.0	16.5	160.0	16.5	7.0

[a] N = Number of samples used to define the mean direction.

[b] Site omitted from Fig. 6 due to uncertain tectonic correction.

components of magnetization generally were isolated after removing a randomly distributed component below 200°C. The component demagnetized between 200 and 450°C is regarded as secondary, and the high-temperature component above 500°C is regarded as the ChRM (Fig. 5b). Again these two components were often similar in orientation, and the secondary component was less well-grouped than the ChRM within a site (Table 3).

The grey limestones had more variable NRM intensities, which ranged from 5×10^{-5} to 4×10^{-3} A/m. A component with inconsistent direction was removed below 150°C during thermal demagnetization. One or two components of magnetization were defined above 150°C. If a secondary component was present in the specimen, it was generally removed between 150 to 330°C. The characteristic component was defined between 330 and 520°C (Fig. 5c). At sites where both the Alba and Barcaliente Formations could be sampled, ChRM directions in the different lithologies were found to be similar (Table 3).

ChRM directions of the Carboniferous limestones are very well-grouped within each site, with α_{95} less than 8° in all but two (AC11 and AC22) sites.

Restoration of pretectonic directions

The ChRM directions in the Asturian arc do not show any significant grouping before tectonic correction (Fig. 6). After applying only a simple tilt correction around a horizontal axis, the dispersion of the direction decreases, with southeasterly directions for Cambrian sites and scattered directions predominantly in the southeast quadrant for Carboniferous sites. However, the complicated geometry of the Cantabrian structures requires that each site be evaluated to see if more elaborate models yield more accurate results. In evaluating the models, it is necessary to decide how to unfold the different superimposed structures that are responsible for the overall structure of an area. A distinction must be made between local restorations at a site, and the general restoration of the arc. Since the restoration of the arc only requires rotations about vertical

axes, the local restorations will be considered first, and the evolution of the arc will be compensated for later. Bonhommet et al. (1981) also found this procedure to be the optimum for correcting the paleomagnetic vectors in their study in the Villasecino area.

Local structural corrections at Carboniferous sites

The local tectonic setting at a site may result from superimposed structures which must be accounted for when making restorations. Structural corrections have traditionally been done in the

TABLE 3

Site mean directions from the Carboniferous Alba and Barcaliente Formations

Site	N [a]	In situ		After local tectonic correction		After removal of radial folds		
		D	I	D	I	D	I	α_{95}
Group I								
AC1	8	182.0	−15.0	187.5	35.0	167.5	35.0	3.2
AC2	8	185.0	−11.5	194.5	35.0	174.5	35.0	2.6
AC3	6	149.0	32.5	162.0	4.0	142.0	4.0	7.8
AC4	10	182.0	−10.0	211.0	35.0	191.0	35.0	5.8
AC5	9	200.0	33.5	182.0	14.0	172.0	14.0	3.3
AC6	9	154.0	−21.0	164.0	29.0	154.0	29.0	3.9
AC7	15	173.0	−2.5	181.0	32.0	171.0	32.0	5.1
AC21	10	203.5	30.0	184.0	4.0	179.0	4.0	2.7
Group II								
AC19	10	117.5	−7.0	129.5	16.5	129.5	16.5	2.5
AC20	11	156.0	18.5	131.5	14.0	131.5	14.0	3.9
Group III								
AC8	9	105.0	59.0	39.0	9.0 [b]			3.9
AC9	8	74.0	−30.0	107.0	21.0	124.0	21.0	4.2
AC10	9	144.0	−15.0	64.5	−50.0 [b]			4.2
AC11	7	60.0	12.5	103.0	20.0	120.0	20.0	12.7
AC12	9	77.0	−19.0	78.0	24.0	95.0	24.0	4.8
AC13	10	153.0	−33.0	157.5	45.5 [b]			3.2
AC14	10	136.5	−50.5	25.0	−22.5 [b]			3.3
AC15	10	117.0	−48.0	33.0	−9.0 [b]			7.9
AC16	9	96.0	6.0	116.0	10.0	130.0	10.0	5.3
AC17	8	69.5	12.0	90.0	14.0	107.0	14.0	4.9
AC18	5	84.0	4.0	87.0	7.0	95.0	7.0	5.5
Group IV								
AC26	13	109.0	14.5	133.0	34.0	133.0	34.0	5.6
AC27	24	180.5	−47.0	327.0	−36.0	327.0	−36.0	3.3
AC28	10	120.0	20.5	155.0	44.0	155.0	44.0	7.8
Group V								
AC22	6	166.0	33.0	150.0	60.0	141.0	60.0	13.0
AC23	9	163.0	−13.5	156.5	43.5	147.5	43.5	7.0
AC24	8	184.0	0.0	184.0	−1.5 [b]			7.4
AC25	9	183.0	−49.0	350.5	−62.5	341.5	−62.5	7.1

[a] N = Number of samples used to define the mean direction.

[b] Site omitted from Fig. 6 due to uncertain tectonic correction.

reverse order from which they were acquired, although the correct order is not always self-evident. Any rock displacement due to tectonic deformation can be described as a rotation. Consequently, a sequence of superimposed deformations can be combined into a single equivalent rotation. For convenience of analysis the paleomagnetic sites in this study were subdivided into five groups based upon geographical distribution (Table 3).

Groups I and II

The sites west of the Central Coal Basin (between the Cantabrian coast and Teberga) belong to a region characterized by folds with different wavelengths and fold axes which are horizontal to gently plunging. No superposition of radial folds is observed in these areas. Two local fold tests were made in this region.

Sites AC5 and AC6 were located on opposite limbs of the same fold. The paleomagnetic directions at the two sites were each Fisher-distributed, the mean directions being: at site AC5, $D = 200°$, $I = 34°$, $\alpha_{95} = 3°$, $N = 9$, and at site AC6, $D = 154°$, $I = -21°$, $\alpha_{95} = 4°$, $N = 9$. These directions are 70° apart. After correcting for a slight plunge of the fold axis, and for the dips of the fold limbs, the mean directions become: at site AC5, $D = 182°$, $I = 14°$, and at site AC6, $D = 164°$, $I = 29°$. These directions are only 22°

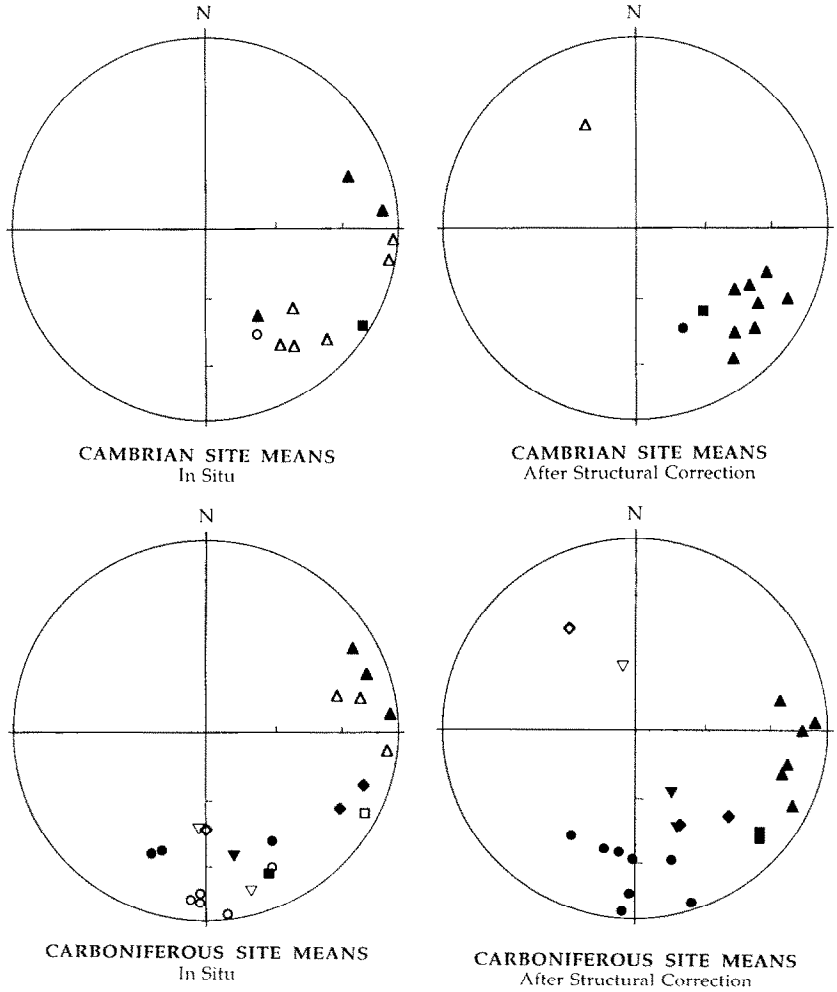


Fig. 6. Equal area plot of site mean directions of the ChRM before and after correction for tectonic tilt. For this and subsequent figures, closed symbols are plotted on the lower hemisphere and open symbols on the upper hemisphere. Circles = Group I; squares = Group II; triangles = Group III; diamonds = Group IV; inverted triangles = Group V.

apart, but this difference is greater than the sum of their individual angles of confidence. The structural corrections dramatically reduce the separation of the site mean directions, but do not result in a common mean direction. The trajectories do not intersect, so there is no evidence that the directions are of synfolding origin. The overall result of the fold test is negative, although the improvement in the directional separation of the site mean directions justifies the interpretation that they are prefolding in origin.

Another fold test was possible for sites AC19 and AC20. Before correction the site mean directions were: at site AC19, $D = 117°$, $I = -7°$, $\alpha_{95} = 3°$, $N = 10$, and at site AC20, $D = 156°$, $I = 19°$, $\alpha_{95} = 4°$, $N = 11$. These directions are 46° apart. The fold axis was horizontal and after simple unfolding for the dips of the fold limbs the mean directions become: at site AC19, $D = 130°$, $I = 17°$, and at site AC20, $D = 132°$, $I = 14°$. These directions are only 3.5° apart, less than the sum of their individual angles of confidence. If the test of McFadden and Jones (1981) is applied at the sample hierarchical level, the value of the test parameter is 0.11, which is less than the critical value of 0.17 for $N = 21$; the directions are not significantly different after tectonic correction. This positive fold test indicates a predeformational origin of the magnetizations at these sites.

Group III

Structures trend east–west in the southern limb of the Asturian arc. The Bodón and Correcilla units consist of straight structures composed of folds with horizontal fold axes, which may plunge at their noses. Sites AC9, AC11, AC12, AC16 and AC17 were taken in these units where the fold axes were horizontal and only a simple tilt correction was required. Site AC9 was corrected additionally for a secondary fold structure. For sites AC8, AC10, AC13, AC14 and AC15, despite using all available structural information, it was not possible to correct the paleomagnetic vector unambiguously.

Group IV

Three sites in this group come from a single fold structure at a locality east of the Central Coal Basin in the Rioseco thrust sheet. This sheet is a detachment slice that is cross-folded by a synform of the radial set. Julivert and Arboleya (1986) unfolded this structure to determine the azimuth and dip of the ramp. The paleomagnetic vectors were corrected using this same procedure. The uncorrected site mean directions were distinctly different (Table 3). After making the multiple structural corrections at each site, the mean direction of sites AC26 and AC28 on one limb of the fold was: $D = 142°$, $I = 39°$, $\alpha_{95} = 5.5°$, $N = 23$. The direction at site AC27 on the other limb had opposite polarity; its antipodal direction was: $D = 147°$, $I = 36°$, $\alpha_{95} = 3.5°$, $N = 24$. The angle between the corrected means is 5°, less than the sum of their angles of confidence. Computed at sample level, the test parameter of McFadden and Jones (1981) is 0.060, marginally less than the critical value of 0.069 for $N = 47$. This positive fold test also indicates a predeformational origin of the magnetizations at these sites.

Group V

Structures on the northern limb of the Asturian arc also trend east–west between Ribadesella and Unquera. At sites AC22, AC23, and AC25 the strike of bedding is east–west, parallel to the regional trend. The fold axes here are presumed to be horizontal. Site 24 was next to a major fault zone, and had low inclinations compared to the other three sites and will be excluded from further interpretation.

Local structural corrections at Cambrian sites

All Cambrian sites were located in the same geographic areas as Groups I, II and III (Table 2). Most of the sites required only a simple tilt-correction. Site AQ10 was in an area where the steep plunge of the fold axis changed from near-vertical to overturned. Since it was not possible to measure the azimuth or dip of the plunge precisely, a unique structural correction was not possible, and this site was not considered further. The paleomagnetic direction at site AQ6 was anomalous before and after structural corrections and was omitted from further analysis.

Sites AQ1, AQ2 and AQ3 are located at the

nose of the Somiedo syncline so that it is necessary to compensate for the plunge of the fold structure. The uncorrected site mean directions lie widely apart on the upper hemisphere. After corrections for fold axis plunge and bedding tilt the site means come close together, although the circle of confidence of site AQ1 does not intersect the circles of AQ2 and AQ3, which overlap. The three sites do not give a perfect fold test, but the marked reduction in dispersion (k increases from 10 to 135), especially in view of the complexity of the structural corrections involved, encourages the interpretation that the remanent magnetization predates the tectonic deformation.

Restoration of the arc

Two types of structures have contributed to the formation of the arc (Julivert and Arboleya, 1984, 1986): (1) thrust sheets emplaced in a rotational manner, and (2) radial folding superimposed on earlier structures. The rotational motion of a thrust sheet can be modelled as the rotation of a subhorizontal plane about a vertical axis. Julivert and Arboleya (1984, 1986) consider the pole of rotation to be located at the point of convergence of an anticline/syncline couple into which motion of the thrust sheet is transferred (cf. Julivert and Arboleya, 1986, fig. 2). Radial folding results in two types of rotation. Firstly, the fold limbs are rotated around the fold axes. Secondly, the arc is tightened by the vanishing of these radial folds towards the convex side of the arc (Julivert, 1971a; Julivert and Marcos, 1973; Julivert and Arboleya, 1984, 1986); this tightening can be modelled as a rotation about a vertical axis.

The Carboniferous vectors are the most useful to reconstruct the evolution of the arc, since sites are located around the entire arc. The Cambrian sites are confined to the southern limb by the limited outcrop exposure. The ChRM directions

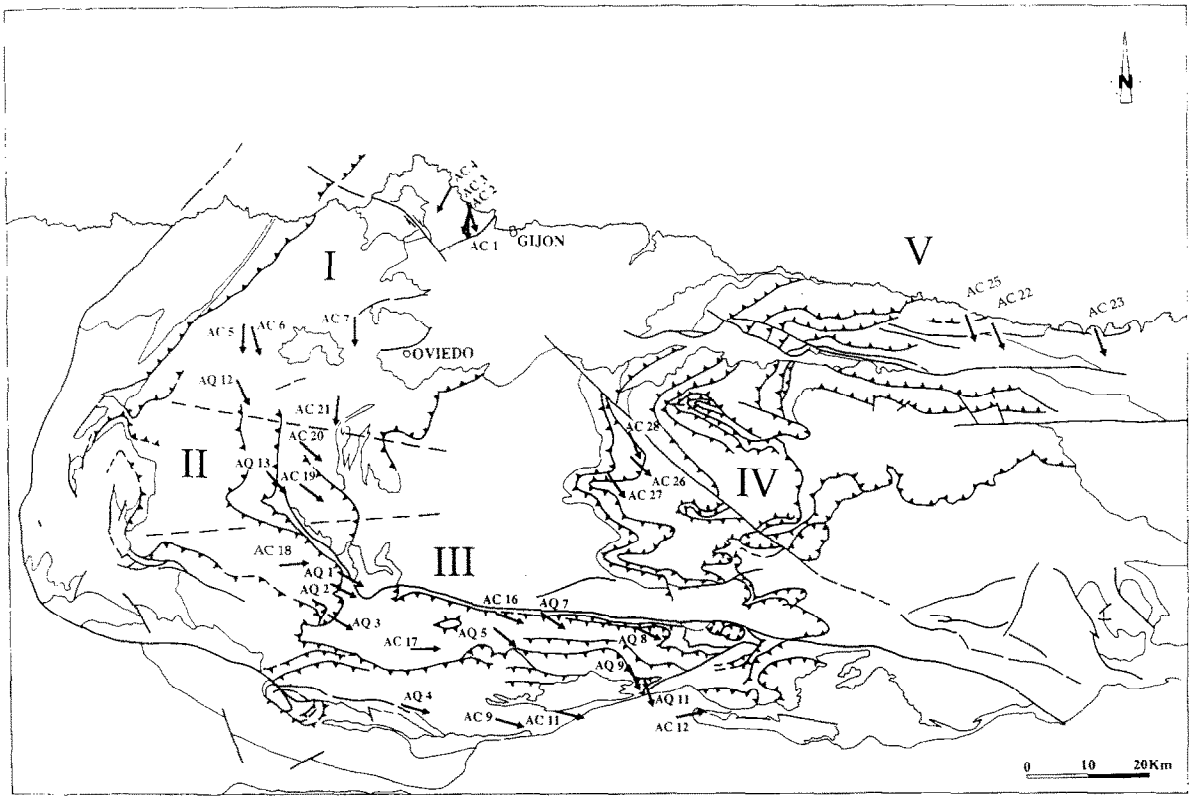


Fig. 7. Paleodeclinations of site means for the Cambrian and Carboniferous sites, after local tectonic corrections. Symbols as in Figure 1.

of the Cambrian and Carboniferous site means after structural correction are shown in Figure 6. The ChRM declinations projected on a map of the arc are shown in Figure 7. In general the declinations lie close to the strike of the arc. The Carboniferous declinations vary around the arc by 135°. They allow five groups of directions to be distinguished, corresponding to the five areas shown in Figure 7.

The declination (D) and inclination (I) of the mean direction of each group, the radius of the circle of 95% confidence of the mean (α_{95}) and the precision parameter (k) which describes the dispersion of the ChRM directions were calculated for each of the 5 groups. The 8 sites in Group I, in the northwestern part of the arc, are characterized by: $D = 182°$, $I = 24°$, $\alpha_{95} = 14°$, $k = 17$. This direction is in agreement with the directions from Perroud (1983) in the Cabo de Peñas. Group II, located in the central part of the arc to the west of the Central Coal Basin, consists of only 2 sites which have $D = 131°$, $I = 15°$. The 6 sites in Group III in the southern limb have an easterly mean direction given by $D = 97°$, $I = 16°$, $\alpha_{95} = 13°$, $k = 29$. This direction is similar to that found by Bonhommet et al. (1981) in the Villasecino anticline which is located at the western limit of the southern limb. Group IV, located in the Rioseco thrust sheet, to the east of the Central Coal Basin, consists of 3 sites with mean direction given by $D = 145°$, $I = 38°$, $\alpha_{95} = 16°$, $k = 63$. This direction is similar to that of Group II, but the Rioseco inclinations are consistently steeper. Three sites form Group V in the northern limb which has a mean direction of $D = 158°$, $I = 56°$, $\alpha_{95} = 18°$, $k = 49$. The inclinations in this group are very steep compared to the other four groups. Two of the sites, AC22 and AC25, which are geographically close together have exactly opposite polarities. It is difficult to explain the steep inclinations of these sites unless they were magnetized after tilting was initiated. However, it was not possible to do a fold test in this area in order to examine for synfolding magnetization.

The mean direction for the nineteen Carboniferous sites is $D = 143°$, $I = 28°$. The directions have a rather large and non-uniform dispersion

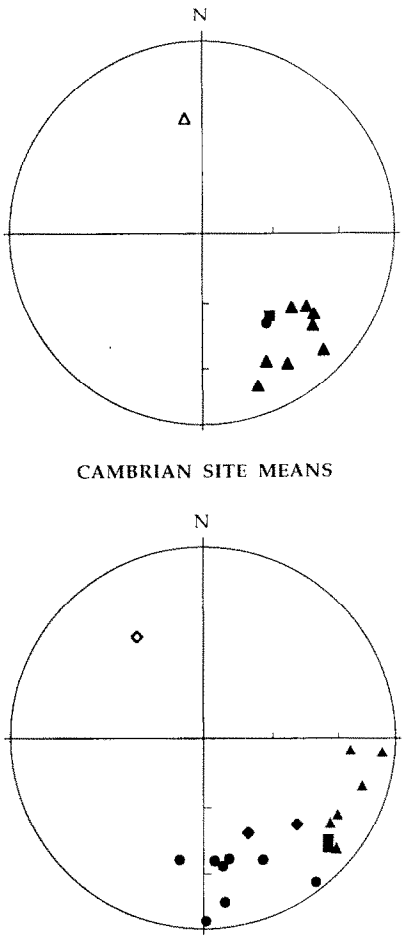


Fig. 8. Equal area plot of the ChRM directions of sites means after removal of the regional radial folds.

which invalidates the application of Fisher statistics. The standard deviations of the declinations (ΔD) and inclinations (ΔI), computed arithmetically, are: $\Delta D = 40°$, $\Delta I = 13°$. The declinations change progressively from north to south around the Asturian arc. This variation includes the effects of the radial folding.

A detailed evaluation of the amount of tightening produced by the radial folding is given by Julivert and Arboleya (1984). Figure 8 and Table 3 show the site mean directions after removing the effects of the radial folding. Each site-mean declination was rotated by an amount predicted from the model of Julivert and Arboleya (1984). This does not greatly alter the overall mean direction ($D = 142°$, $I = 25°$). It improves the grouping of the paleomagnetic vectors although

the declinations are still dispersed ($\Delta D = 28°$). This may imply rotation of thrust sheets during their emplacement.

The ChRM directions at 11 Cambrian sites (Fig. 8) are non-uniform azimuthally before removing the effects of the radial folding, and have mean direction $D = 131°$, $I = 31°$ with $\Delta D = 26°$, $\Delta I = 10°$. Most of the sites were limited to the southern limb, but removal of rotations due to the radial folding reduces the dispersion of the directions so that Fisher-statistics can be applied. The corrected mean direction is $D = 142°$, $I = 31°$, $\alpha_{95} = 9°$, $k = 25$, $\Delta D = 13°$, $\Delta I = 10°$.

Discussion

The similarity between the Cambrian and Carboniferous directions suggests that the Cambrian samples may have been remagnetized in the Carboniferous, before the initiation of folding. It is not possible to evaluate this point adequately because of the lack of suitable reference directions for Cambrian rocks from Spain. However, the Hercynian orogeny in northern Spain advanced from the west (Galicia) towards the east. By analogy with the Alleghenian orogeny in the American Appalachian fold belt, the deformation in Asturias may have been accompanied by fluid migration that resulted in remagnetization during a precursory phase of the deformation. It is also conceivable that the Carboniferous (Visean and Namurian) rocks were remagnetized prior to the initiation of folding in the Westphalian. Opdyke (1991) has inferred from a magnetostratigraphic study of the Mauch Chunk Formation in the Appalachian fold belt that the reversal rate in the Visean and Namurian was about one reversal per million years. With two exceptions all our Carboniferous sites have reverse polarity. By implication from Opdyke's results, if these sites are remagnetized, the remagnetization probably took place relatively quickly.

Fold tests were carried out on structures in the Asturian arc where sites could be placed on opposite limbs of the same fold. Only one of the four fold tests was decisively positive. The lack of statistical significance in the other three fold tests may due to a more complicated fold geometry than could be accounted for by field observations. In each of these other tests the dispersion between sites was minimum at 100% unfolding. Also, the grouping of all Carboniferous site mean directions improves markedly when tectonic corrections are applied. We conclude that ChRM was acquired in the Carboniferous rocks prior to folding. For the purposes of this study the Carboniferous and Cambrian samples can be assumed to carry a pre-folding magnetization.

The declinations of the Carboniferous sites change progressively around the curve of the fold belt. However, it is important to note that the angle between the declinations and the strike of the folds is not constant. In the south the declinations are eastward, and make a small angle with the E–W strike of the folds. In the north, the declinations are more or less southwards, and make a large acute angle with the SW–NE trend of the fold axes.

A summary of Late Paleozoic paleomagnetic poles for Europe (Van der Voo, 1990) predicts that Late Carboniferous/Early Permian directions for the Asturias should have $D = 7°$, $I = -21°$ (reversed direction, $D = 187°$, $I = +21°$). Middle Carboniferous to Late Devonian directions should have $D = 9°$, $I = -39°$ (reversed direction $D = 189°$, $I = +39°$). If 35° of anticlockwise rotation of Iberia in the Cretaceous are taken into account, the expected reversed directions for our Carboniferous sites should be around $D = 150°–155°$, $I = 20°–40°$. If Iberia were associated with Gondwana, the corresponding reversed Carboniferous directions for Spain would be $D = 133°–142°$, $I = 10°–30°$ (Van der Voo, pers. commun., 1991). These conditions are approximated most closely by the Group II sites in the west ($D = 131°$, $I = 15°$), where the fold axes strike nearly north–south.

Assuming that this area has undergone no rotation, the northern area with the Group I sites must have been rotated clockwise and the southern area containing the Group III sites has been rotated anticlockwise. This observation allows us to discriminate between the several models proposed for the origin of the curvature of the Asturian arc. For example, it contradicts the interpretation of Ries and Shackleton (1976) that anti-

clockwise rotation should have occurred over the Asturian portion of the Ibero-Armorican arc, and that the feature is secondary in origin. At least part of the observed curvature is primary. Our results also do not support the model suggested by Pérez-Estaún et al. (1988), which predicts that the outer thrust sheets should show progressively greater clockwise rotation around the arc. It is compatible with the models that imply clockwise rotation of the northern end of the arc and anticlockwise rotation of the southern end (Matte and Ribeiro, 1975; Julivert and Arboleya, 1984, 1986; Matte 1986).

After compensating for rotations associated with the radial folds, the arc trends approximately 50° for Group I and 305° for Group III; the difference is 105°. The rotated mean paleomagnetic declinations of Group I (168°) and Group III (113°) still show a significant residual difference of 55° (Fig. 9a). If the mean declinations of Groups I and III are aligned with the mean declination of Group II, the arc becomes straighter but does not lose its curvature (Fig. 9b); the residual difference in trend is 50°. This difference is greater than the sum of the declination errors (estimated by $\alpha_{95}/\cos I$) for Group I (15°) and Group III (14°). This implies that the arc was originally not linear, but possessed a small primary curvature that may mirror the shape of the margin of the depositional basin at the time of deposition of these sediments.

Conclusions

The evident relationship between paleomagnetic directions and curvature of the fold belt implies that paleomagnetic data from the Asturian arc should be omitted from compilations used in the construction of a Paleozoic apparent polar wander path.

The Asturian arc is now tightly curved. Part of the curvature may have been present before the further evolution of the arc, which proceeded in two stages. The first stage involved rotations of thrust sheets during their emplacement (Julivert and Arboleya, 1986), producing a more curved arc than the original form. In a second stage, the development of radial folds further tightened the

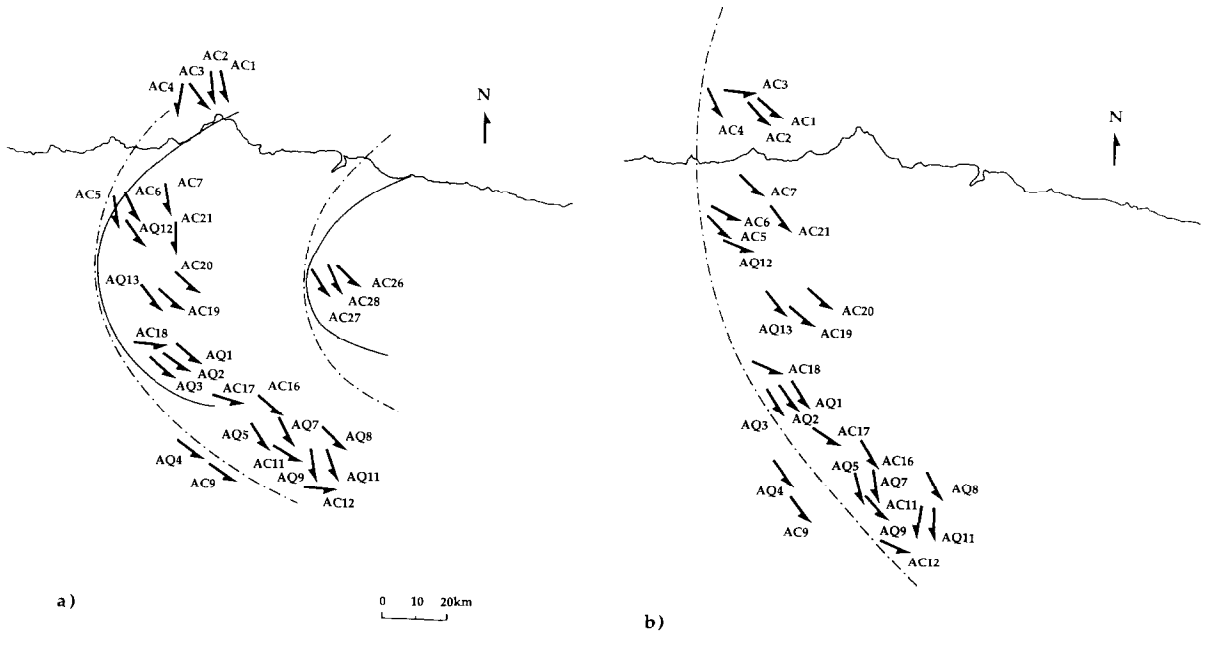


Fig. 9. (a) Interpretative model showing the effects of correcting declinations and strikes of fold axes for removal of the radial folds. Solid line marks present trace of the arc; dashed line shows the trace of the arc after correcting for radial folds. (b) The dashed line shows the trace of the arc after aligning the paleomagnetic declinations of group means to a common declination of 150°.

curvature of the arc. Both stages would lead to clockwise rotations in the north and anticlockwise rotations in the south.

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